

AYE
TECH HUB

• ELECTRICAL ENGINEERING • LESSON 06

Electrical Tools

Multimeter · Insulation Tester · Loop Impedance · PAT

Lesson 06 of 30 | Electrical Engineering Program

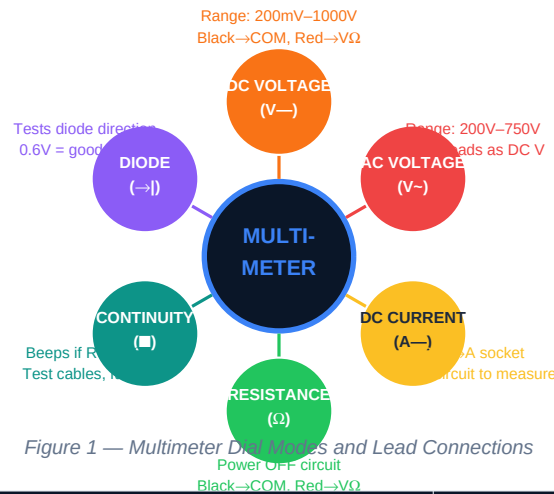
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- The multimeter — every mode explained with step-by-step procedure
- Insulation resistance tester — why 500V and 1000V tests exist
- Loop impedance tester — verifying protective device operation
- PAT testing — what it is, what it checks, what it cannot check
- Safe instrument use — CAT ratings and lead inspection before every test

The Multimeter — Your Most Essential Electrical Tool

The digital multimeter (DMM) is the single most important tool for any electrician or electrical engineer. It measures voltage, current, resistance, continuity, and more. **Every electrical fault-finding session starts with a multimeter.** Used correctly it is safe. Used incorrectly — wrong range, wrong leads, wrong CAT rating — it can be lethal.



Measurement	Procedure	Common Mistake to Avoid
DC Voltage	1. Set dial to DC V (V—). Select range above expected voltage. 2. Black lead → COM. Red lead → VΩ terminal. 3. Touch probes to circuit. Read display.	Never measure voltage with leads in the current (A) socket — creates a near short circuit and blows the fuse or destroys the meter
AC Voltage	1. Set dial to AC V (V~). Select 750V range for mains. 2. Same lead positions as DC voltage. 3. Touch probes to live and neutral terminals.	Never select DC voltage range when measuring AC — display may show near zero even though dangerous live voltage is present
Resistance	1. Power OFF the circuit completely. Isolate and verify dead. 2. Set dial to Ω. Select appropriate range. 3. Touch probes across component. Read ohms.	Never measure resistance on a live circuit — damages meter and gives false reading
Continuity	1. Set dial to continuity (diode symbol). Power circuit off. 2. Touch probes to both ends of wire/fuse/switch. 3. Audible beep = continuous. Silence = open circuit.	Continuity beep does not mean the circuit is safe — only that there is a low-resistance path between the two probe points
DC Current	1. Set dial to DC A. Move red lead to A socket. 2. Break the circuit — meter goes IN SERIES with the load. 3. Use clamp meter for high currents — safer and no need to break circuit.	Connecting ammeter in parallel (like a voltmeter) creates a short circuit — destroys the meter and can cause arc flash

Before every use: Inspect the test leads. Cracked or exposed insulation, corroded tips, or damaged plugs are dangerous — replace them. Check the instrument's CAT rating matches your work environment. A CAT II meter is not safe for panel work — use CAT III minimum.

Insulation Resistance, Loop Impedance and Earth Testing

Beyond the multimeter, professional electrical work requires specialist test instruments that cannot be improvised. These are the tools that verify an installation is safe before it is energised — or that find faults after installation problems develop.

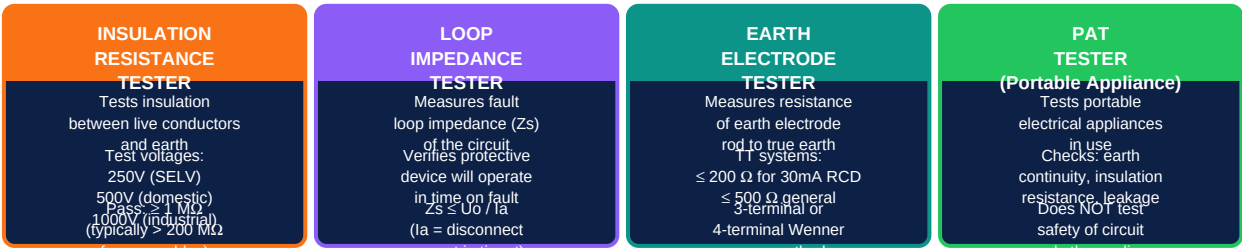


Figure 2 — Electrical Test Instruments: Insulation / Loop / Earth / PAT

Insulation Resistance Testing — Procedure

Step	Action	Why
1— Isolate and disconnect	Isolate circuit. Disconnect all equipment (lamps, appliances, sensitive electronics). Short L+N together at the board.	Test voltage can damage electronic equipment. Shorting L+N tests both conductors at once.
2— Select test voltage	250V for SELV/PELV circuits. 500V for 230V domestic/commercial. 1000V for 400V industrial circuits.	Higher voltage stresses the insulation — reveals degraded insulation that lower voltages miss.
3— Connect test leads	One lead to L+N (shorted). Other lead to PE (earth). Apply test voltage.	Measures resistance of insulation between conductors and earth.
4— Read and record	New installation: $\geq 200\text{ M}\Omega$ (pass). Minimum acceptable: $\geq 1\text{ M}\Omega$ per IEC 60364.	Readings below $1\text{ M}\Omega$ indicate damaged, wet, or contaminated insulation.
5— Reconnect and re-energise	Remove shorting link. Reconnect equipment. Restore supply. Verify operation.	Normal sequence after any insulation test.

Loop Impedance Testing

Test	What Is Measured	Pass Criterion	Standard
Z_e — External earth fault loop impedance	Impedance of the supply network earth loop, measured at the origin (consumer unit)	Depends on supply type — typically $< 0.35 \Omega$ for TN-S	BS 7671 / IEC 60364
Z_s — Total earth fault loop impedance	Full fault loop: supply + live conductor + protective conductor. Measured at socket or end of circuit.	$Z_s \leq U_o / I_a$ where I_a = current to trip MCB in required time (e.g. 32A Type B MCB: $I_a = 160A \rightarrow Z_s \leq 230/160 = 1.44 \Omega$)	BS 7671 Table 41.3-41.5
$R1 + R2$ — Resistance of live + protective conductors	Combined resistance of circuit live conductor + CPC (circuit protective conductor)	Used to calculate $Z_s = Z_e + (R1+R2)$. Low = good: shorter cable or larger CSA.	IEE On-Site Guide

Why loop impedance matters: If Z_s is too high, the fault current during an earth fault will be too low to trip the MCB within the permitted disconnection time (0.4 sec for socket circuits, 5 sec for fixed). This means live metalwork could remain energised long enough to kill. Z_s testing proves the protection system will work.

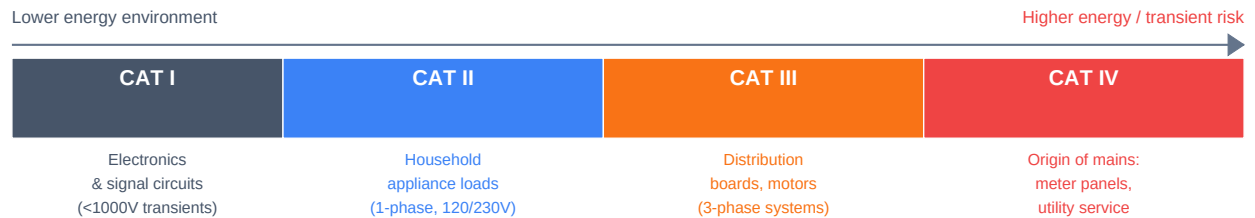
PAT Testing, CAT Ratings and Hand Tools

PAT testing (Portable Appliance Testing) checks that portable electrical equipment is safe to use. It is a mandatory part of workplace electrical safety management in many countries. Importantly, PAT testing checks the **appliance** — not the circuit it is plugged into. Both need separate verification.

PAT Test	What It Checks	Pass Values	Instruments Used
Visual inspection (always first)	Damaged plug, frayed cable, broken housing, signs of overheating, incorrect fuse	No visible damage, correct fuse for appliance rating	Eyes only — the most important part of PAT
Earth continuity	Resistance of earth conductor from plug pin to exposed metal parts of the appliance	Class I appliances: $< 0.1 \Omega$ + cable resistance Extended leads: resistance increases with length	PAT tester or low-resistance ohmmeter
Insulation resistance	Resistance between live conductors and earth/case at 500V DC	Class I: $\geq 1 \text{ M}\Omega$ Class II (double insulated): $\geq 2 \text{ M}\Omega$	PAT tester or insulation resistance tester
Leakage current (substitute test)	Current flowing from live parts through insulation to earth during operation at mains voltage	Class I: $\leq 3.5 \text{ mA}$ (0.75 mA preferred)	PAT tester with leakage current function
Functional check	Does the appliance actually work correctly when energised?	Operates as designed — no sparks, overheating, or abnormal behaviour	Operate the appliance under supervision

Instrument CAT Ratings — Never Ignore Them

Figure 3 — Instrument CAT Rating: Always Use $\geq$ CAT III for Panel Work



Essential Hand Tools for Electrical Work

Tool	Purpose	Safety Requirement
Insulated screwdrivers (IEC 60900)	Tighten/loosen terminals in live panels and equipment	Rated 1000V. VDE mark confirms genuine insulation. Never use unrated screwdrivers near live parts.
Voltage indicator (non-contact)	Detect presence of AC voltage without contact — pen-type detector	Test on known live before use to verify it works. Never rely on it alone — always verify with multimeter.
Cable stripper	Remove insulation from cable cores without nicking the conductor	Always strip cables de-energised. Use correct gauge stripper for cable size.
Crimping tool	Create permanent mechanical and electrical connections on terminals	Use correct die for terminal size. Under- or over-crimp fails the connection.
Clamp meter (current clamp)	Measure AC current without breaking the circuit — clamp around ONE conductor	CAT III or IV minimum. Only ONE conductor in the jaw — both conductors cancel out.
Phase rotation tester	Verify correct L1-L2-L3 phase sequence before connecting 3-phase motors	Reverse rotation destroys pumps and compressors — always test before energising.

What comes next — ELEC-007: Electrical Wiring Basics — cable types and selection (PVC, LSZH, armoured, flexible), conductor sizing by current rating and voltage drop, ring main vs radial circuits, and reading wiring diagrams.